		$\sqrt{\dfrac{\quad}{5}} = 340\ \text{m s}^{-1}$ (2 s.f.)
15	**C**	$ma = mg - N$ $ma = mg \quad \text{if } N = 0$ $m\omega^2 x_0 = mg$ $x_0 = \dfrac{g}{\omega^2} = \dfrac{9.81}{(2\pi f)^2} = \dfrac{9.81}{4\pi^2 (2.0)^2} = 6.2\,\text{cm}$ (2 s.f.)
16	**A**	Graph shows U vs r of object undergoing simple harmonic motion. Hence $F = -kr$